User Acceptance Testing (UAT) Template

Date	22 February 2026
Team ID	LTVIP2026TMIDS42033
Project Name	**Weather-Based Prediction of Wind Turbine Energy Output: A Next-Generation Approach to Renewable Energy Management**
Maximum Marks	

Project Overview:

Project Name: Weather-Based Prediction of Wind Turbine Energy Output:
A Next-Generation Approach to Renewable Energy Management

Project Description: This project integrates real-time weather data with a machine learning regression model to predict wind turbine energy output. The system supports renewable energy management by providing accurate forecasting, dashboard analytics, and decision-support insights.

Project Version: 1.0

Testing Period: 20 February 2026 to 25 February 2026

Testing Scope:

Features Tested:

- Weather data fetching (API integration)
- Energy prediction using ML model
- Dashboard visualization
- Filters (City, Wind Speed, Date)
- Efficiency calculation
- Backend API response validation

Testing Environment:

URL/Location:
Frontend – http://localhost:5173
Backend – http://127.0.0.1:5000

Test Cases:

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-001	Fetch Weather Data	Enter city → Click search	Weather data displayed	Data displayed correctly	Pass
TC-002	Predict Energy Output	Enter weather values → Click Predict	Predicted energy (kW) shown	Prediction displayed	Pass
TC-003	Dashboard Filters	Apply wind speed filter	Graph updates dynamically	Filter works correctly	Pass
TC-004	Efficiency Calculation	Check efficiency %	Correct percentage calculated	Calculation correct	Pass
TC-005	Invalid City Input	Enter wrong city	Error message displayed	Error shown correctly	Pass
TC-006	Backend Offline Test	Stop backend → Predict	Show backend error message	Error displayed	Pass

Bug Tracking:

Bug ID	Bug Description	Steps to Reproduce	Severity	Status	Additional Feedback
BG-001	Weather API delay	Slow network → Fetch weather	Low	Closed	API timeout optimized
BG-002	Minor UI alignment issue	Resize screen	Low	Closed	Responsive fix applied

Sign-off:

Tester Name:

Date:

Signature:

Notes:

- Ensure that all test cases cover both positive and negative scenarios.
- Encourage testers to provide detailed feedback, including any suggestions for improvement.
- Bug tracking should include details such as severity, status, and steps to reproduce.
- Obtain sign-off from both the project manager and product owner before proceeding with deployment.